

ELIZABETHTOWN, NC, USA

WMO: 720279

Lat: 34.602N

Lon: 78.578W

Elev: 133

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	27.6	5.0	7.1	32.9	9.7	9.1	36.7	19.6	51.8	17.4	55.4	2.5	320	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	95.1	76.2	92.9	75.6	90.9	74.9	79.1	88.6	78.2	87.0	77.4	85.6	6.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	141.9	80.8	76.2	137.5	80.5	75.2	133.0	80.1	42.5	88.9	41.6	87.8	40.8	85.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.9	11.2	17.7	98.7	3.2	1.9	15.4	100.0	13.5	101.1	11.7	102.2	9.3	103.6
			15.4	80.8	3.1	1.2	13.2	81.6	11.4	82.3	9.6	83.0	7.4	83.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	45.6	47.7	54.6	63.5	71.3	78.5	81.0	79.9	74.6	65.5	53.9	49.4	(6)
	DBStd	14.74	10.43	9.20	9.40	7.66	6.30	4.67	3.62	3.87	5.20	7.65	8.52	9.71	(7)
	HDD50	613	215	142	63	5	0	0	0	0	0	2	53	133	(8)
	HDD65	2498	604	486	341	118	21	0	0	0	2	90	344	492	(9)
	CDD50	5680	78	79	205	409	659	856	961	927	739	481	171	115	(10)
	CDD65	2091	3	3	18	72	215	406	496	462	292	104	12	8	(11)
	CDH74	18599	7	27	169	621	1775	3839	4951	4220	2205	693	66	26	(12)
	CDH80	7368	0	1	18	133	598	1662	2233	1792	785	140	6	0	(13)
(14) Wind	WSAvg	3.8	4.3	4.8	5.1	5.0	4.0	3.7	3.5	2.6	2.3	3.0	3.2	3.9	(14)
(15) Precipitation	PrecAvg	49.1	3.7	3.1	3.7	3.2	3.6	4.9	5.4	6.2	5.6	3.5	3.3	3.1	(15)
	PrecMax	63.5	7.2	6.8	7.2	5.9	6.6	12.5	11.5	12.3	17.5	7.4	9.6	5.8	(16)
	PrecMin	33.0	1.1	1.3	0.6	0.6	1.3	0.8	2.2	2.4	0.9	0.4	0.3	0.7	(17)
	PrecStd	7.5	1.9	1.5	1.6	1.5	1.4	2.7	2.2	2.5	3.9	2.2	2.2	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.8	78.0	82.3	86.8	94.5	97.0	97.5	97.0	93.0	87.8	79.5	77.4	(19)
		MCWB	64.5	63.6	62.8	65.6	72.5	75.8	76.3	76.3	72.7	72.4	65.3	67.9	(20)
	2%	DB	70.4	72.3	78.9	83.9	89.2	94.1	95.2	93.7	90.6	83.9	75.4	72.6	(21)
		MCWB	62.1	60.5	63.0	65.6	70.1	75.7	76.7	76.7	73.7	70.4	64.9	66.1	(22)
	5%	DB	66.4	68.1	74.8	81.1	86.2	91.3	92.8	91.2	87.9	81.1	71.9	69.7	(23)
		MCWB	60.0	57.7	60.5	64.3	69.8	74.3	76.3	76.0	73.1	68.8	62.8	64.9	(24)
	10%	DB	62.9	63.7	70.5	77.5	83.0	89.0	90.6	89.3	85.0	78.2	68.5	65.8	(25)
		MCWB	57.2	55.5	58.9	63.2	69.1	73.4	75.7	75.5	72.5	68.0	61.9	62.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.6	66.7	67.8	73.3	75.8	79.4	80.4	80.2	78.3	76.2	71.0	71.1	(27)
		MCDB	72.0	73.4	74.5	78.3	84.2	90.2	90.1	89.1	84.7	80.6	75.9	74.2	(28)
	2%	WB	64.2	63.5	66.0	70.8	73.8	78.2	79.2	78.8	77.2	74.4	68.5	68.4	(29)
		MCDB	68.3	68.5	73.4	76.9	81.7	88.0	89.0	87.7	84.0	79.8	71.6	71.3	(30)
	5%	WB	61.4	60.3	64.1	68.3	72.6	77.0	78.2	78.1	76.0	72.7	66.0	65.6	(31)
		MCDB	65.4	66.2	70.5	75.0	80.5	86.0	87.4	86.4	82.4	77.5	69.6	68.3	(32)
	10%	WB	58.1	57.1	62.0	66.3	71.4	75.9	77.3	77.3	74.9	70.7	63.2	62.1	(33)
		MCDB	62.2	62.4	68.2	73.8	79.1	84.3	86.1	85.0	80.7	75.4	66.9	65.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	20.5	21.9	22.3	20.2	18.9	18.5	17.4	17.7	19.2	20.9	19.1	(35)
		MCDBR	22.2	24.5	25.6	25.3	22.8	20.9	20.1	19.7	20.9	20.7	22.8	20.5	(36)
	5% WB	MCWBR	15.3	15.2	12.1	10.7	7.6	6.6	6.1	6.1	6.9	9.3	13.4	14.2	(37)
		MCWBR	20.0	22.5	21.0	20.0	18.4	19.1	18.4	17.4	15.9	15.7	18.3	18.5	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.322	0.332	0.367	0.403	0.465	0.497	0.540	0.516	0.431	0.390	0.345	0.321	(40)
		taud	2.493	2.458	2.387	2.281	2.154	2.110	2.009	2.067	2.303	2.384	2.466	2.529	(41)
	Ebn at Noon	Ebn at Noon	281	290	287	280	263	253	242	245	263	265	269	274	(42)
		Edh at Noon	27	31	36	42	48	50	56	51	39	33	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	860	1069	1413	1767	1924	2007	1919	1716	1463	1221	945	756	(44)
	RadStd	RadStd	51	106	107	114	148	125	122	93	116	142	75	61	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+218

Nomenclature: See separate page